Unit 9: Magnetic Effect of Electric Current

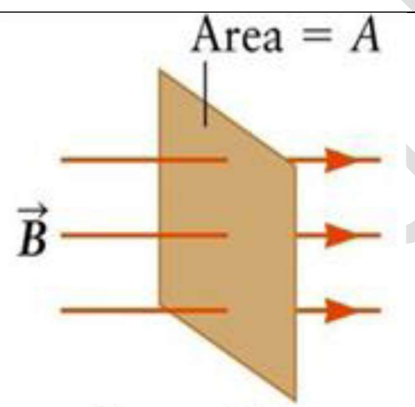
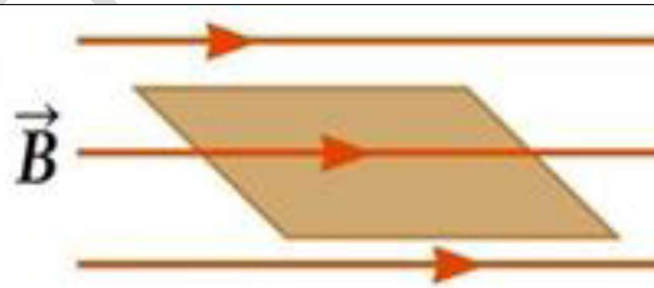
Lesson 1: Meaning of Magnetic Field

General formula:

$$\Phi = B\ A \cos(\ \Theta\ )$$

Where:

- **Φ**: is the magnetic flux through a surface (**Weber = Wb**)
- B: the magnetic flux density (**Tesla**)
- A: the surface area (**m^2**)
- Θ : the angle between the perpendicular to (normal to) surface and direction of magnetic field (**degrees**)

	Maximum magnetic flux	Minimum magnetic flux
Diagram		
Theta (Θ)	Zero	90°
Cosine	1	Zero
Formula	$\Phi = B\ A$	$\Phi = Zero$

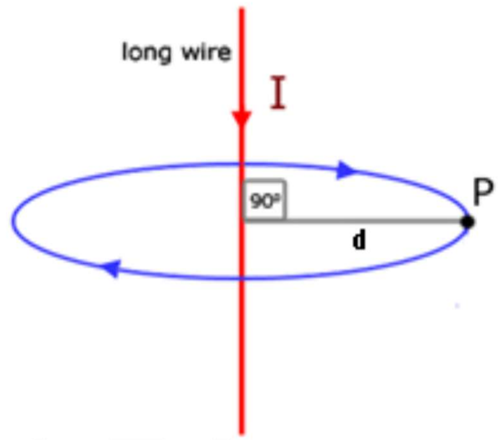
Lesson 2: Magnetic field due to current passing through a straight conductor

General formula (Ampere's circular law):

$$B = \frac{\mu I}{2\ \pi\ d}$$

Where:

- B: the magnetic field density (**Tesla**).
- I: intensity of current passing through the wire (**Ampere**).
- d: The distance of the point from the wire (**meter**).
- μ : magnetic Permeability of medium. (**Wb/m.A = Tesla.m/A**)



Note: Permeability of air: $4\pi * 10^{-7}$

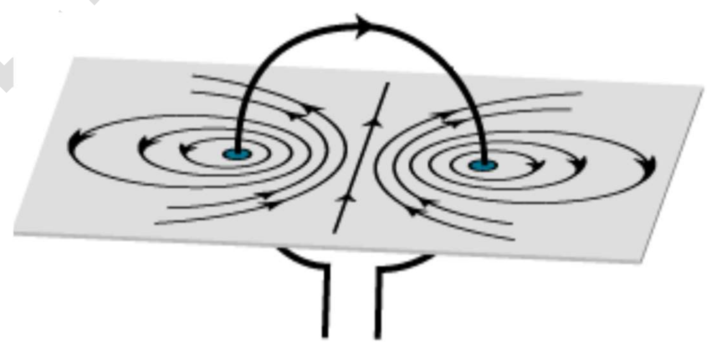
Lesson 3: Magnetic field due to current passing through a circular coil at its center

General formula (at the center of the circular loop):

$$B = \mu \frac{N\,I}{2\,r}$$

Where:

- B: the magnetic field density (**Tesla**).
- N: The number of turns
- I: intensity of current passing through the wire (**Ampere**).
- r: the radius of the circular coil (meter)
- μ : magnetic Permeability of medium. (**Wb/m.A = Tesla.m/A**)



Note: Permeability of air: $4\pi * 10^{-7}$

Wire ➔ Circular coil

If a straight wire of length (L) is bent to form circular coil of radius (r) and of turn's number (N):

$$L = 2\,\pi\,r\,N$$

Where:

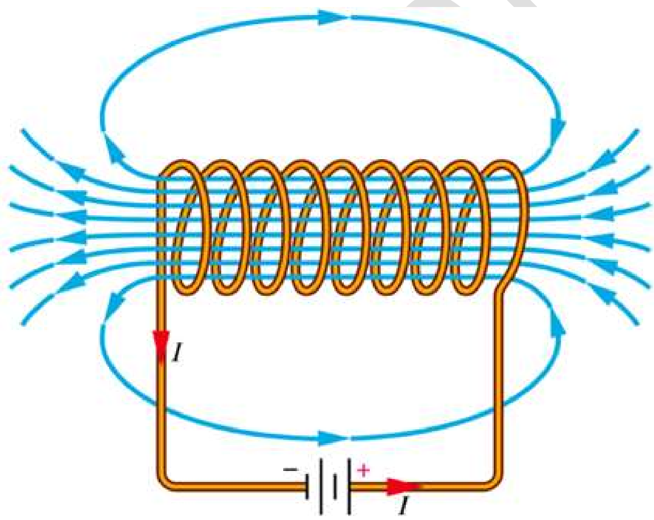
- L: circumference of the circular coil or length of the straight wire (**meter**)

- r: the radius of the circular coil (**meter**)
- N: The number of turns

Lesson 4: magnetic field due to current passing through a solenoid

General formula (at a point on its interior axis):

$$B = \mu \frac{NI}{L}$$



Where:

- B: the magnetic field density at the center of the circular coil (**Tesla**)
- I: intensity of current passing through the coil (**Ampere**).
- L: the length of the solenoid (**meter**).
- μ: Permeability of medium ($\mu_{air} = 4\pi\times10^{-7}$ **Wb/m.A.**)
- N: the number of turns of wire in the solenoid

$$B = \mu n I$$

- n: number of turns per unit length n = N/L (**turns/meter**)

Lesson 5: the magnetic force acting on a wire carrying current placed in a uniform magnetic field

General formula:

$$F = B\,I\,L \sin(\Theta)$$

Where:

- F: force of the wire (**Newton N**)
- L: length of the wire (**meter m**)
- I: current in the wire (**ampere A**)
- B: magnetic flux density (**Tesla T**)

- Θ : angle between the wire (direction of current) and the magnetic flux density (**degrees**)

	Maximum force	Minimum force
Theta (Θ)	90° or 270°	Zero or 180°
Sine	1	Zero
Formula	$F = B\ I\ L$	$F = Zero$

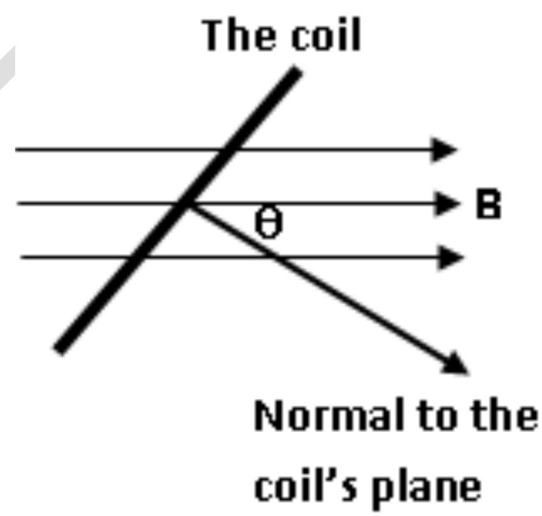
Lesson 6: The magnetic torque acting on a coil carrying current placed in a uniform magnetic field

Magnetic torque formula:

$$\tau = B\ I\ \mathrm{A\ N} \sin(\Theta)$$

Where:

- $\boldsymbol{\tau}$: magnetic torque (**N.m = T.A.m^2**)
- B: magnetic flux density (**Tesla**)
- I: current in the wire (**ampere**)
- A: the area of the coil's plane (**meter square m^2**)
- N: number of turns of the coil
- θ : is the angle between the normal (perpendicular) to coil's plane and direction of magnetic flux lines



The magnetic dipole moment formula:

$$|md| = \frac{\tau_{max}}{B} = I\ \mathrm{A\ N}$$

Where:

- $|\boldsymbol{md}|$: Magnetic dipole moment (**A.m^2 = N.m/ T**)
- $\boldsymbol{\tau}$: magnetic torque (**N.m = T.A.m^2**)

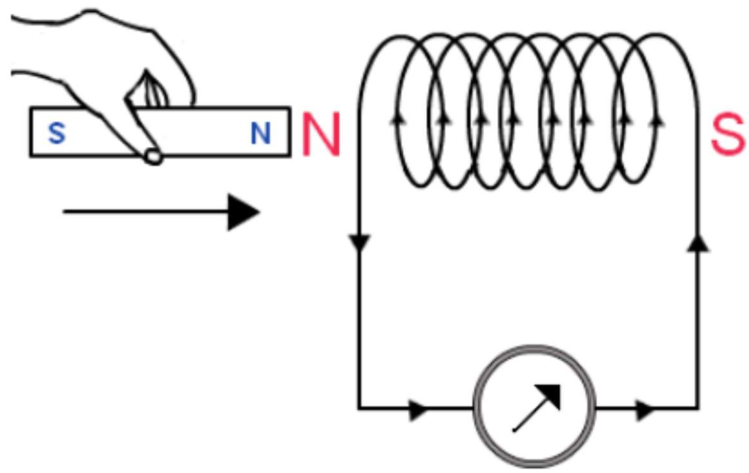
- I: current in the wire (**ampere**)
- A: the area of the coil's plane (**meter square m^2**)
- N: number of turns of the coil

Unit 10: Electromagnetic Induction

Lesson 1: Meaning of electromagnetic induction and faraday's law

Faraday's equation:

$$EMF_{ind} = \frac{-N\,\Delta\Phi_m}{\Delta t}$$



Where:

- $\boldsymbol{EMF_{ind}}$: induced electromotive force (**Volt**)
- N: the number of turns of wire in the solenoid
- $\boldsymbol{\Delta\Phi_m}$: the variation in the magnetic flux intercepted by the conductor through the time interval Δt (**Weber**)
- Δt: time interval (**second**)

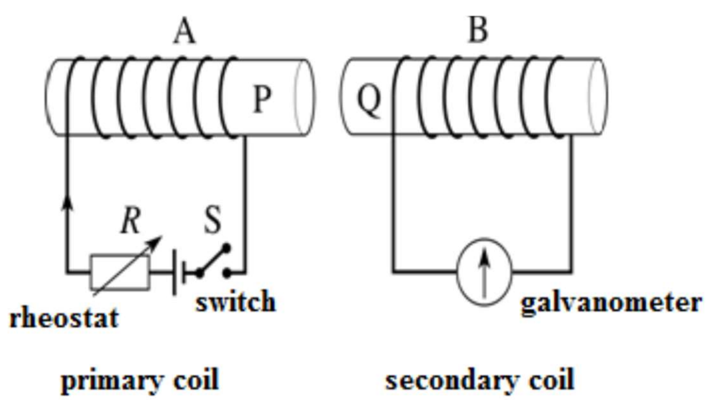
Note: The negative sign in the above relation indicates that the direction of the induced electromotive force or the induced current tends to oppose the cause producing it. This rule is known as Lenz's rule.

Lesson 3: Mutual and self-induction

Mutual induction:

$$emf_2 = -M\,\frac{\Delta I_1}{\Delta t} = -N_2\,\frac{\Delta\Phi_2}{\Delta t}$$

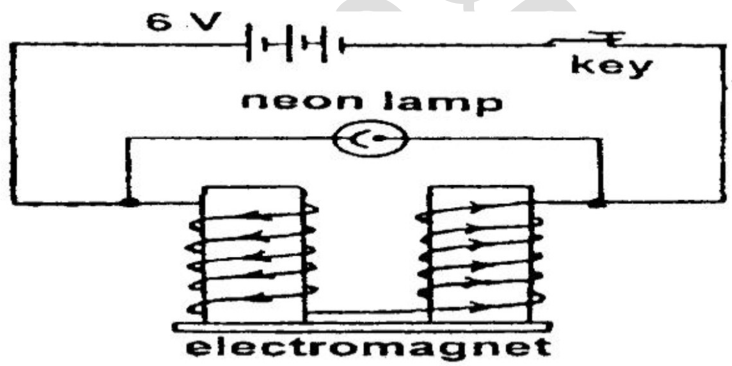
Where:



- $\boldsymbol{emf_2}$: The induced emf in the secondary coil (volt)
- M: Coefficient of mutual induction (mutual inductance) (Henry)
- $\frac{\Delta I_1}{\Delta t}$: The rate of change of current in the primary coil (ampere/sec)
- $\boldsymbol{N_2}$: Number of turns of the secondary coil
- $\frac{\Delta \Phi_2}{\Delta t}$: The rate of change of magnetic flux linking with the secondary coil (Weber/sec)

Self-induction:

$$emf = -L\,\frac{\Delta I}{\Delta t} = -N\,\frac{\Delta \Phi}{\Delta t}$$



Where:

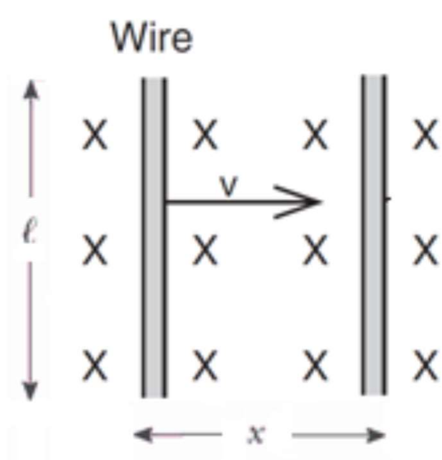
- emf: The induced emf in the secondary coil (volt)
- L: coefficient of self-induction (self-inductance) (Henry)
- $\frac{\Delta I}{\Delta t}$: The rate of change of current in the coil (ampere/sec)
- N: the number of turns of coil
- $\frac{\Delta \Phi}{\Delta t}$: The rate of change of magnetic flux in the coil (weber/sec)

Lesson 4 The electric generator (dynamo)

The induced emf in a straight wire moving in a magnetic field:

$$emf = B\,l\,v\,sin(\theta)$$

Where:

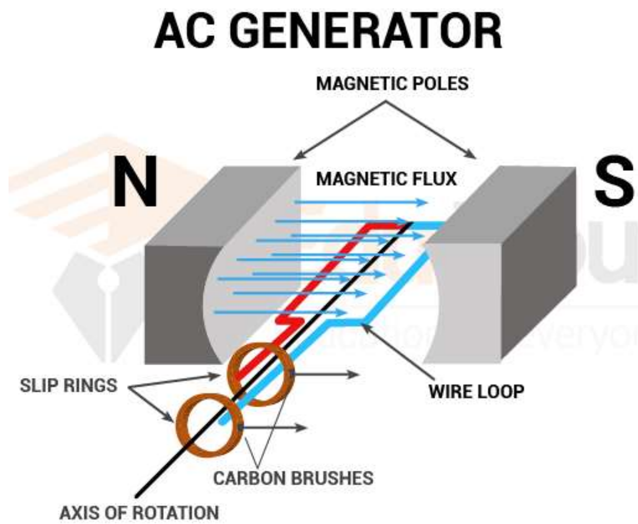


- emf: induced electromotive force (volt)
- B: magnetic field of density (Tesla)
- L: length of the wire (meter)
- V: velocity of the wire (meter/second)
- $\boldsymbol{\theta}$: angle between B and V (degrees)

The induced EMF in a dynamo:

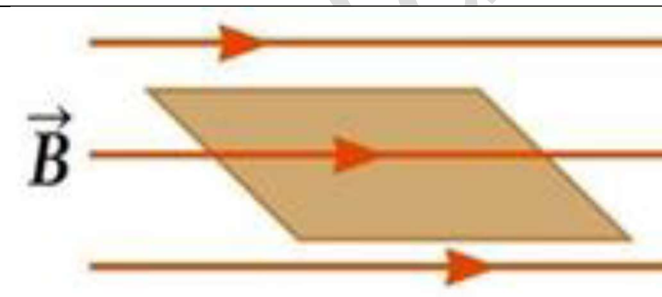
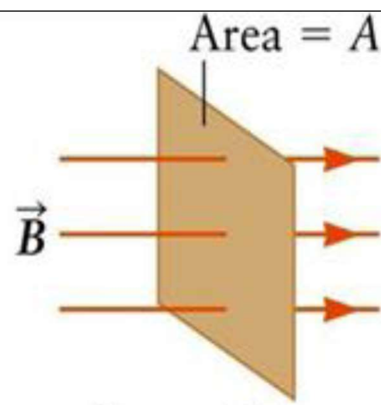
$$emf_{coil} = B\ A\ N\ (2\ \pi\ f)\ sin(\theta)$$

$$emf_{coil} = B\ A\ N\ (2\ \pi\ f)\ sin(2\ \ \pi\ f\ \ t)$$



Where:

- $\boldsymbol{emf_{coil}}$: represents the instantaneous value of induced EMF (**volt**)
- B: uniform magnetic field (**Tesla**)
- A: area of the rectangular coil (**meter square**)
- N: number of turns
- F: frequency (rotating the coil around its axis) (**hertz**)
- $\boldsymbol{\theta}$: the angle between normal on coil's plane (axis of the coil) and magnetic flux lines. (**degrees**)
- $\boldsymbol{t}$: time (**seconds**)

	Maximum emf	Minimum emf
Diagram		
Theta (Θ)	90°	Zero
Sine	1	Zero
Formula	$emf_{coil} = B\ A\ N\ (2\ \pi\ f)\ sin(\theta)$	$emf_{coil} = Zero$

The induced Current in a dynamo:

$$I = \frac{emf_{coil}}{R}$$

Where:

- emf_{coil}: represents the instantaneous value of induced EMF (volt)
- I: instantaneous value of induced current I generated in the coil (Ampere)
- R: total resistance (ohm)

Lesson 5: The electric transformer

Formulas of ideal electric transformer:

$$P_p = P_s$$

Where:

- P_p: power at primary coil (Watts)
- P_s: power at secondary coil (Watts)

$$V_p * I_p = V_s * I_s$$

Where:

- V_p: voltage (potential difference) at primary coil (Volts)
- V_s: voltage (potential difference) at secondary coil (Volts)
- I_p: current at primary coil (Ampere)
- I_s: current at secondary coil (Ampere)

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K$$

Where:

- N_p: Number of turns of primary coil
- N_s: Number of turns of secondary coil
- K: transformer ratio (equal to 1 in ideal transformer)

Efficiency of a non-ideal transformer:

$$\eta = \frac{output\ power}{input\ power} * 100$$

$$\eta = \frac{P_s}{P_p} * 100$$

$$\eta = \frac{V_s * I_s}{V_p * I_p} * 100$$

$$\frac{\eta\, V_p}{V_s} = \frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Unit 11: Electromagnetic Induction

Lesson 1: black body radiation phenomenon

Planck's distribution:

$$E = h\, f = h\, \frac{c}{\lambda}$$

Where:

- $\boldsymbol{E}$: Energy of a photon (Joules)
- $\boldsymbol{h}$: Plank's constant (6.626×10^{-34} joule/hertz or joule*seconds)
- $\boldsymbol{f}$: Photon's frequency (Hertz)
- $\boldsymbol{c}$: Speed of light ($3*10^8$ meter/second)
- $\boldsymbol{\lambda}$: Wavelength (meter)

Note: You can get plank's constant from the calculator by

1. Shift + 7
2. 06

Wien's Law:

$$\lambda_{max} * T = constant$$

Where:

- $\boldsymbol{\lambda_{max}}$: Wavelength at maximum radiation intensity (meter)
- $\boldsymbol{T}$: Temperature (Kelvin)

- ***constant***: YOU have to CALCULATE it each time.

Note: If temperature was given in Celsius, you will use K = 273 + C to convert it to kelvin.

Lesson 2: Thermionic Emission

CRT formula:

$$K.E. = eV = \frac{1}{2} m_e v^2$$

Where:

- $\boldsymbol{K.E.}$: Kinetic energy of the incident electron (Joules)
- $\boldsymbol{e}$: electron charge ($1.6*10^{-19}$ C)
- $\boldsymbol{V}$: potential difference between anode and cathode (Volts)
- $\boldsymbol{m_e}$: mass of the electron ($9.109*10^{-31}$ kg)
- $\boldsymbol{v}$: electron velocity (m/s)

Note: You can use the calculator to get

1. Shift + 7
2. 03 ➔ mass of electron ($\boldsymbol{m_e}$)
3. 23 ➔ electron charge (e)

Lesson 3: Photoelectric Effect

Formula:

$$K.E._{max} = E - E_w$$

$$K.E._{max} = hf - hf_{threshold}$$

Where:

- $\boldsymbol{K.E._{max}}$: Kinetic energy of the emitted electron (Joules)
- $\boldsymbol{E}$: Energy of a photon (Joules)
- $\boldsymbol{E_w}$: Work function of the metal (Joules)
- $\boldsymbol{h}$: Plank's constant (6.626×10^{-34} joule/hertz or joule*seconds)

- f: Photon's frequency (Hertz)
- $f_{threshold}$: the lowest frequency below which the photoelectric effect does not occur. (Hertz)

Lesson 4:

Conservation of momentum:

Total linear momentum before collision = total linear momentum after collision

Conservation of energy:

Sum of energy of photon and electron before collision = sum of energy of photon and electron after collision.

Momentum of photon:

$$P_L = \frac{E}{c} = \frac{h\,f}{c} = \frac{h}{\lambda}$$

Where:

- P_L: Momentum of photon (kg.m/s)
- E: Energy of a photon (Joules)
- h: Plank's constant (6.626×10^{-34} joule/hertz or joule*seconds)
- f: Photon's frequency (Hertz)
- c: Speed of light ($3*10^8$ meter/second)
- λ: Wavelength (meter)

Lesson 5: X–Rays

In continuous radiation:

$$eV = h\,f_{max} = h\,\frac{c}{\lambda_{min}}$$

Unit 12: Modern electronics

Lesson 4: Transistor

Formulas:

$$I_E = I_C + I_B$$

$$I_C = \alpha_e \, I_E$$

$$I_B = (1 - \alpha_e) \, I_E$$

$$\beta = \frac{I_C}{I_B} = \frac{\alpha_e \, I_E}{(1 - \alpha_e) \, I_E} = \frac{\alpha_e}{1 - \alpha_e}$$

Where:

- $\boldsymbol{I_E}$: emitter current (Ampere)
- $\boldsymbol{I_C}$: collector current (Ampere)
- $\boldsymbol{I_B}$: base current (Ampere)
- $\boldsymbol{\alpha_e}$: percentage of electrons diffusing across the junction
- $\boldsymbol{\beta}$: current gain

THE END